

T. Lee



**TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2001**

# **MATHEMATICS**

## **EXTENSION II**

*Time Allowed – 3 Hours  
(Plus 5 minutes reading time)*

**All questions may be attempted**

**All questions are of equal value**

**In every question, show all necessary working**

**Marks may not be awarded for careless or badly arranged work**

**Standard integral tables are printed at the end of the examination paper and may be removed for your convenience. Approved silent calculators may be used.**

**The answers to all questions are to be returned in separate bundles clearly labelled Question 1, Question 2, etc. Each bundle must show your candidate number.**

**Question 1****Marks**

- (a) Find  $\int \frac{x^3}{x-2} dx$ . 2
- (b) Evaluate  $\int_{-2}^2 (x \cos^2 x - 100x^5 + 2) dx$ . 2
- (c) (i) Express  $\frac{3x+7}{(x+1)(x+2)(x+3)}$  in partial fractions. 3
- (ii) Prove that  $\int_0^1 \frac{(3x+7)dx}{(x+1)(x+2)(x+3)} = \ln 2$ . 2
- (d) Let  $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$  where  $n$  is a non negative integer.
- (i) Show that  $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x dx$  where  $n \geq 2$ . 3
- (ii) Deduce that  $I_n = \frac{n-1}{n} I_{n-2}$  where  $n \geq 2$ . 2
- (iii) Evaluate  $I_4$ . 1

**Question 2 (Start a new page)**

- (a) The hyperbola  $h$  has equation  $x^2 - y^2 = 4$ .
- (i) Find the foci, asymptotes and vertices. 3
- (ii) Prove that the equation of the normal to  $h$  at  $P(4, 2\sqrt{3})$  is  $2\sqrt{3}y + 3x = 24$ . 3
- (iii) Find the equation of the circle that is tangent to  $h$  at  $P$  and  $Q(-4, 2\sqrt{3})$ . 3
- (iv) Using complex numbers or otherwise show that  $x^2 - y^2 = 4$  transforms to  $xy = 2$  if we choose the asymptotes as the  $x$  and  $y$  axes with the curve in the first and third quadrants. 3
- (v) Find the foci of  $xy = 2$ . 1
- (b) Of three cards, one is green on both faces, one white on both faces, whilst the third is green on one side and white on the other. They are placed in a hat, one is withdrawn and placed on a table. If the visible face is green, what is the probability that the other face is also green? 2

**Question 3 (Start a new page)**

### Marks

- (a) Define modulus and conjugate of a complex number  $z = x + iy$  ( $x, y$  real). Prove that:

1

- (i)  $|z|^2 = z \bar{z}$  and that, for any two complex numbers  $z_1$  and  $z_2$ .

**2**

- $$(ii) \quad (\overline{z_1 z_2}) = (\overline{z_1}) (\overline{z_2}).$$

**2**

- (iii) Deduce that  $|z_1 z_2| = |z_1| |z_2|$ .

**2**

- (b) Draw neat, labeled sketches (not on graph paper) to indicate each of the subsets of the Argand diagram described below (all necessary detail such as intercepts to be shown)**

- (i)  $\{z: 1 \leq |z| \leq 3 \text{ and } 0 \leq \arg z \leq \frac{\pi}{2}\}.$

**2**

- (ii)  $\{z : |z + 1| + |z - 1| = 3\}$ .

**3**

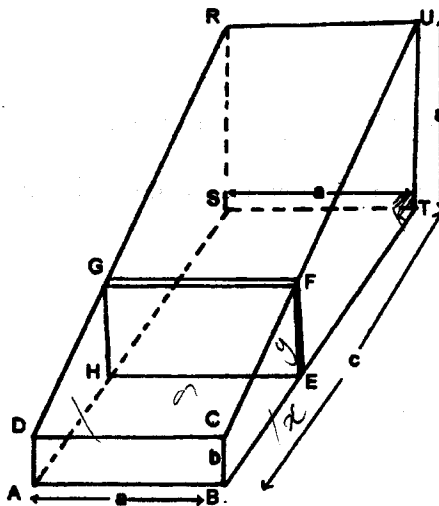
- (iii)  $\{z : \arg(z-2) - \arg(z+2) = \frac{\pi}{3}\}.$

3

### Question 4 (Start a new page)

(a)

5



The diagram shows a solid with rectangular base ABTS. The end ABCD is a rectangle, and the other end STRU is a square. Both ends are perpendicular to the base.

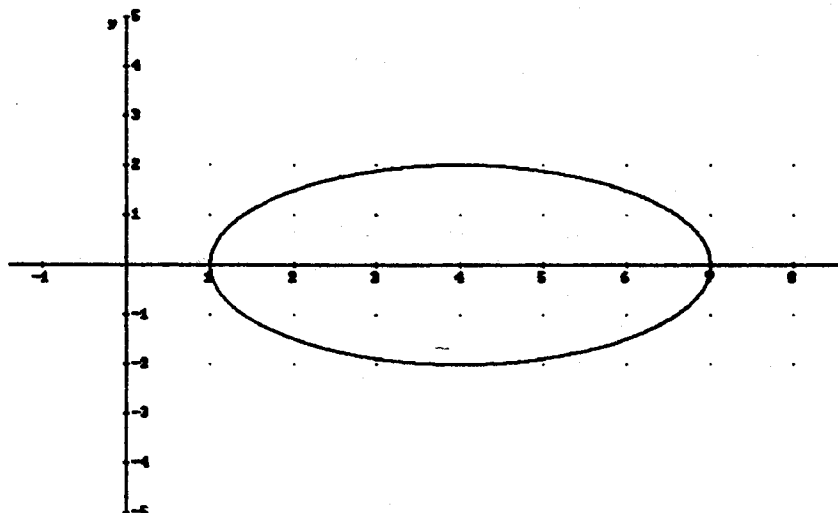
**Consider the slice of the solid with face HEFG and thickness  $\Delta x$  metres, and  $BE=AH=x$  metres.**

- (i) Show that the cross sectional area of this slice is  $\frac{a}{c}[bc + (a - b)x]$ .

- (ii) Hence find the volume of the solid.**

- (b) The ellipse  $\frac{(x-4)^2}{9} + \frac{y^2}{4} = 1$  is rotated about the y axis forming a doughnut shape.

6



- (i) By taking slices perpendicular to the axis of rotation show that the volume of a slice is  $8\pi\sqrt{36-9y^2}\delta y$ .
- (ii) Find the volume of the solid.
- (c) Prove by induction that for every natural number  $n$ , if  $A_1, A_2, \dots, A_n$  are pairwise distinct points, no three of which are on one line, then these points determine exactly  $\frac{n(n-1)}{2}$  lines.

4

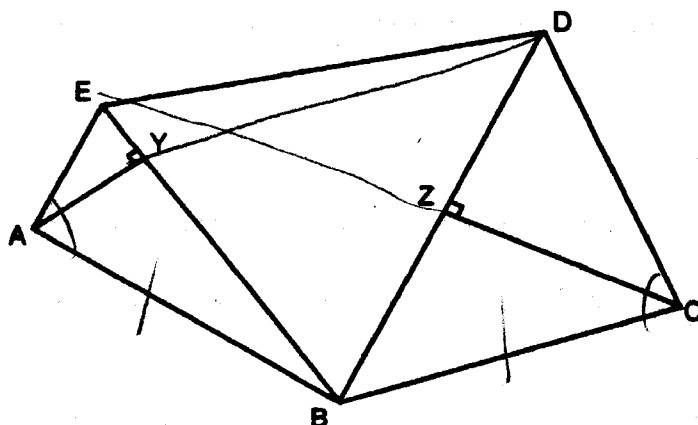
**Question 5 (Start a new page)**

**Marks**

(a) Let  $f(x) = \sin x$ ,  $-\pi \leq x \leq \pi$ . Provide separate sketches of the graphs of:-

- |                        |   |
|------------------------|---|
| (i) $y =  f(x) $ .     | 2 |
| (ii) $ y  = f(x)$ .    | 2 |
| (iii) $ y  =  f(x) $ . | 2 |
| (iv) $y = f( x )$ .    | 2 |

(b) 7



ABCDE is a convex polygon such that  $AB=BC$ ,  $\angle BCD = \angle EAB = 90^\circ$ .  $AY \perp EB$ ,  $BD \perp CZ$

- By using similar triangles prove that  $AB^2 = BY \cdot BE$ .
- Hence prove  $\triangle BEZ \parallel \triangle BDY$ . (You may assume  $BC^2 = BZ \cdot BD$ )
- Show that  $DEYZ$  is a cyclic quadrilateral.

**Question 6 (Start a new page)**

**Marks**

(a) A boat is moving with constant speed in a circle of radius 60m. It does a complete circuit in 1 minute. The total mass of the boat is 300kg and the total resistance it meets is 600 Newtons.

- |                                                                                                                          |   |
|--------------------------------------------------------------------------------------------------------------------------|---|
| (i) Show that $\theta$ , the angle made by the force (F) driving the boat and the tangent, is approximately $18^\circ$ . | 3 |
| (ii) Hence find the force (F) driving the boat.                                                                          | 2 |

(b) A train of mass  $m$ , pulled by a locomotive which exerts a constant (propelling) force  $P$  is moving at speed  $v$  along a straight level track against a resistive force  $mkv$ , where  $k$  is a positive constant. Show that if the speed increases from 2m/s to 4m/s over a time interval of 5 seconds,

- |                                                                                                                         |   |
|-------------------------------------------------------------------------------------------------------------------------|---|
| (i) $P = 2km \left( \frac{2e^{5k} - 1}{e^{5k} - 1} \right)$ .                                                           | 4 |
| (ii) Find the corresponding distance moved.                                                                             | 3 |
| (iii) Prove that there is an upper bound to the speed that the train can attain and find the value of this upper bound. | 3 |

**Question 7 (Start a new page)**

- (a)  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  lie on the parabola  $x^2 = 4ay$  (chord PQ has positive slope)
- (i) If the chord PQ subtends a right angle at the vertex of the parabola show that  $pq = -4$ . 2
- (ii) If the line PQ is inclined at an angle  $\theta$  to the axis of the parabola, show that  $\cot \theta = \frac{p+q}{2}$ . 2
- (iii) Prove that the length of the chord is  $4a \operatorname{cosec} \theta \sqrt{3 + \operatorname{cosec}^2 \theta}$  4
- (b) Let  $z = x + iy$  be a complex number ( $x$  and  $y$  real) satisfying  $z\bar{z} + (1-2i)z + (1+2i)\bar{z} \leq 3$
- (i) Express the inequality in the Cartesian form. 2
- (ii) Sketch the locus of  $z$  on an Argand diagram. 1
- (iii) Find the maximum and minimum values of  $x + y$  by considering lines of the form  $x + y = k$ . 4

**Question 8 (Start a new page)**

- (a) Prove  $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$ . 2
- (b) In  $\triangle ABC$ , angle  $A$  is twice angle  $B$ , angle  $C$  is obtuse and the three lengths  $a, b, c$  are integers.
- (i) Using the sine rule and the formulae for  $\sin 2\beta$  and  $\sin 3\beta$  show that  $a^2 = b(b+c)$  4
- (ii) Show  $2\cos \beta = \frac{a}{b}$ . 3
- (iii) If  $\frac{n}{m} = \frac{a}{b}$  (where  $n$  and  $m$  have no common factor other than 1) deduce that  $b = km^2$  and  $b+c = kn^2$ . 2
- (iv) Show that  $\frac{\sqrt{3}}{2} < \cos \beta < 1$ , and deduce that  $m \geq 4$  and  $n \geq 7$ . 2
- (v) Find the minimum perimeter of  $\triangle ABC$ . 2

**END OF PAPER**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$

$$(a) \int \frac{x^3}{x-2} dx$$

$$= \int \left( x^2 + 2x + 4 + \frac{8}{x-2} \right) dx$$

$$\frac{x^3}{3} + x^2 + 4x + 8 \ln|x-2| + c$$

$$(b) \int_{-2}^2 (x \cos^2 x - 100x^5 + 2) dx$$

$$= [2x]_{-2}^2$$

$$= 8$$

NOTE  $x \cos^2 x$  &  
 $100x^5$  are  
odd

$$\Rightarrow (i) \frac{3x+7}{(x+1)(x+2)(x+3)} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x+3}$$

$$= \frac{A(x+2)(x+3) + B(x+1)(x+3) + C(x+1)(x+2)}{(x+1)(x+2)(x+3)}$$

$$\text{Let } x = -2, x = -3, x = -1$$

$$\therefore B = -1, C = -1, A = 2$$

$$\therefore = \frac{2}{x+1} - \frac{1}{x+2} - \frac{1}{x+3}$$

$$\therefore \int_0^1 \frac{3x+7}{(x+1)(x+2)(x+3)} = \left[ 2 \ln(x+1) - \ln(x+2) - \ln(x+3) \right]_0^1$$

$$= (2 \ln 2 - \ln 3 - \ln 4) - (-\ln 2 - \ln 3)$$

$$= \ln 2$$

$$(d) (i) I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$= \int_0^{\frac{\pi}{2}} \sin^{n-1} x \sin x dx$$

$$= \left[ -\cos x \sin^{n-1} x \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \cos^2 x \sin^{n-2} x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \cos^2 x \sin^{n-2} x dx$$

$$(ii) = n-1 \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) \sin^{n-2} x dx$$

$$= n-1 \int_0^{\frac{\pi}{2}} (\sin^{n-2} x - \sin^n x) dx$$

$$I_n = n-1 I_{n-2} - n-1 \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$I_n + (n-1) I_n = n-1 I_{n-2}$$

$$I_n = \frac{n-1}{n} I_{n-2}$$

$$(iii) I_4 = \frac{3}{4} I_2$$

$$= \frac{3}{4} \left[ \frac{1}{2} I_0 \right]$$

$$= \frac{3\pi}{16}$$

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$I_0 = \int_0^{\frac{\pi}{2}} 1 dx$$

$$= [x]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2}$$

2 (a)  $x^2 - y^2 = 4$

(i)  $b^2 = a^2(e^2 - 1)$   
 $4 = 4(e^2 - 1)$   
 $e = \sqrt{2}$

foci  $(\pm 2\sqrt{2}, 0)$   
 asymptotes  $y = \pm x$   
 vertices  $(\pm 2, 0)$

(ii)  $2x - 2y \frac{dy}{dx} = 0$   
 $\frac{dy}{dx} = \frac{x}{y}$   
 $= \frac{2}{\sqrt{3}}$

$\therefore$  eqn<sup>n</sup> Normal  
 $y - 2\sqrt{3} = -\frac{\sqrt{3}}{2}(x - 4)$   
 $2y - 4\sqrt{3} = -\sqrt{3}x + 4\sqrt{3}$   
 $3x + 2\sqrt{3}y = 24$

(iii) Centre lies on normal  
 & y axis  
 $2\sqrt{3}y + 0 = 24$   
 $y = \frac{24}{2\sqrt{3}}$

$\therefore$  Centre  $(0, 4\sqrt{3})$   
 Radius  $\sqrt{4^2 + (2\sqrt{3})^2}$   
 $= \sqrt{28}$

$\therefore$  Circle  $x^2 + (y - 4\sqrt{3})^2 = 28$

(iv) Let new coords of  $(u, v)$  be  $(X, Y)$

$X + iY = \frac{1}{\sqrt{2}}(1 + i)(x + iy)$   
 $= \frac{x - y}{\sqrt{2}} + i\left(\frac{x + y}{\sqrt{2}}\right)$

$XY = \frac{x^2 - y^2}{2}$

$= 2$  since  $x^2 - y^2 = 4$   
 $\therefore \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right)(2\sqrt{2} + 0i)$

$\frac{1}{2}P(2 \text{ green if known then is 1 green}) = \frac{1/3}{1/2}$   
 $= \frac{2}{3}$

a  $|z| = \sqrt{x^2 + y^2}$   $\bar{z} = x - iy$

(i)  $|z|^2 = (\sqrt{x^2 + y^2})^2 = x^2 + y^2$

$z\bar{z} = (x + iy)(x - iy) = x^2 - (iy)^2$   
 $= x^2 + y^2$

$\therefore |z|^2 = z\bar{z}$

(ii)  $\overline{z_1 z_2} = \overline{(x_1 + iy_1)(x_2 + iy_2)}$

$= x_1 x_2 - y_1 y_2 + i(x_1 y_2 + x_2 y_1)$

$= x_1 x_2 - y_1 y_2 - i(x_1 y_2 + x_2 y_1)$

$= (x_1 - iy_1)(x_2 - iy_2)$

$= \bar{z}_1 \bar{z}_2$

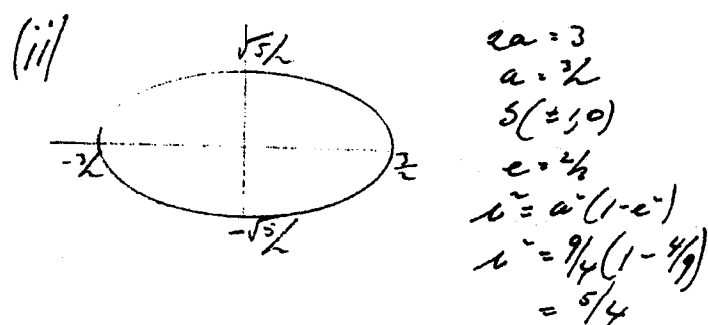
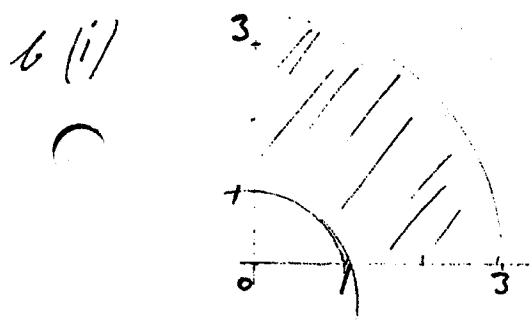
(iii)  $|z_1 z_2|^2 = z_1 z_2 \cdot \overline{z_1 z_2}$

$= z_1 z_2 \cdot \bar{z}_1 \bar{z}_2$

$= z_1 \bar{z}_1 \cdot z_2 \bar{z}_2$

$= |z_1|^2 \cdot |z_2|^2$

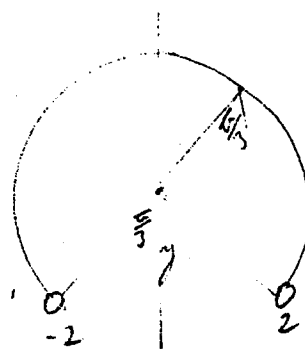
$\therefore |z_1 z_2| = |z_1| |z_2|$



$\therefore \frac{4x^2}{9} + \frac{4y^2}{5} = 1$

$20x^2 + 36y^2 = 45$

(iii)



$\tan \frac{\pi}{3} = \frac{2}{y}$

$y = \frac{2\sqrt{3}}{3}$

center  $(0, \frac{2\sqrt{3}}{3})$

$r^2 = (\frac{2\sqrt{3}}{3})^2 + 4$

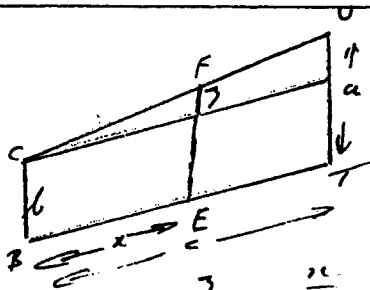
$= \frac{4}{3}$

$\therefore$  are part of circle

$x^2 + (y - \frac{2\sqrt{3}}{3})^2 = \frac{16}{3}$

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(i)



$$\frac{x}{a-b} = \frac{a}{c}$$

$$x = \frac{a(a-b)}{c}$$

$\therefore$  Cross Sectional Area =  $a, b + \frac{x(a-b)}{c}$

OR.

$$FE = b + x \cdot \frac{a-b}{c}$$

$$\text{When } FE = a, x = c$$

$$\therefore b = a - \frac{a-b}{c} \cdot c$$

$$\therefore FE = b + \frac{a-b}{c} \cdot x$$

$$\therefore \text{Cross Sectional Area} = a, b + \frac{x(a-b)}{c}$$

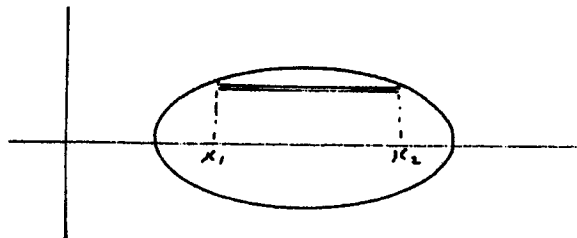
$$(ii) V = \int_0^c \left( b + \frac{x(a-b)}{c} \right) dx$$

$$= \frac{a}{c} \int_0^c (bc + ax - bx) dx$$

$$= \frac{a}{c} \left[ bck + (a-b) \frac{x^2}{2} \right]_0^c$$

$$= abc + \frac{(a-b)ac}{2}$$

b. (i)



$$V_{\text{SLICE}} = \pi (x_2^2 - x_1^2) dy$$

$$= \pi (x_2 - x_1)(x_2 + x_1) dy$$

Now  $x_2$  &  $x_1$  will be the roots

$$\text{of } 4(x-u)^2 + 9v^2 = 36$$

$$4x^2 - 32x + 9y^2 + 28 = 0$$

$$x_1 + x_2 = \frac{32}{4} = 8$$

$$x_2 - x_1 = \sqrt{(x_2 + x_1)^2 - 4x_1x_2}$$

$$= \sqrt{(x_2 + x_1)^2 - 4x_1x_2}$$

$$= \sqrt{64 - 4 \cdot \frac{9y^2 + 28}{4}}$$

$$= \sqrt{36 - 9y^2}$$

$$\therefore V_{\text{SLICE}} = \pi \cdot 8 \sqrt{36 - 9y^2} dy$$

$$(ii) V = \pi \int_{-2}^2 8 \sqrt{36 - 9y^2} dy$$

$$= 24\pi \int_{-2}^2 \sqrt{4 - y^2} dy$$

$$= 48\pi$$

$$(c) \text{ If } n=2 \text{ we obtain } \frac{2(2-1)}{2} = 1$$

which is true.

Assume true for  $n=k$   
i.e. assume that for  $k$  points  
we obtain  $\frac{k(k-1)}{2}$  lines

Prove true for  $n=k+1$   
i.e. prove if we have  $k+1$  points  
we obtain  $\frac{(k+1)k}{2}$  lines

[Note the  $k+1$ th point will  
create an additional  $k$  lines]

$$\therefore \text{ for } n=k+1$$

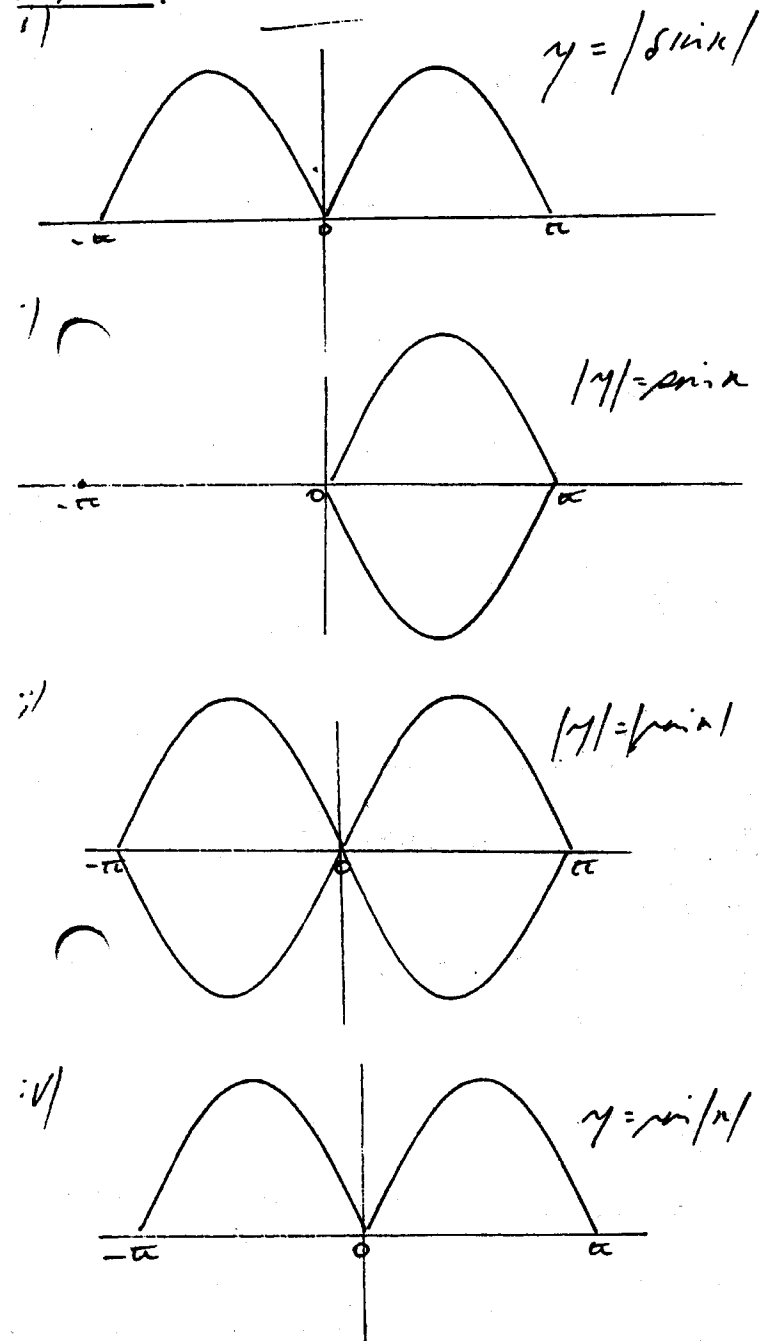
$$\frac{k(k-1)}{2} + k \text{ lines (by assumption)}$$

$$= \frac{k(k-1) + 2k}{2}$$

c (cont)

Thus if it is true for  $k$  terms  
it is true for  $k+1$  terms  
It is true for  $n=2$  & hence  $n=3$   
etc  $\therefore$  true for all  $n$ .

question 5.



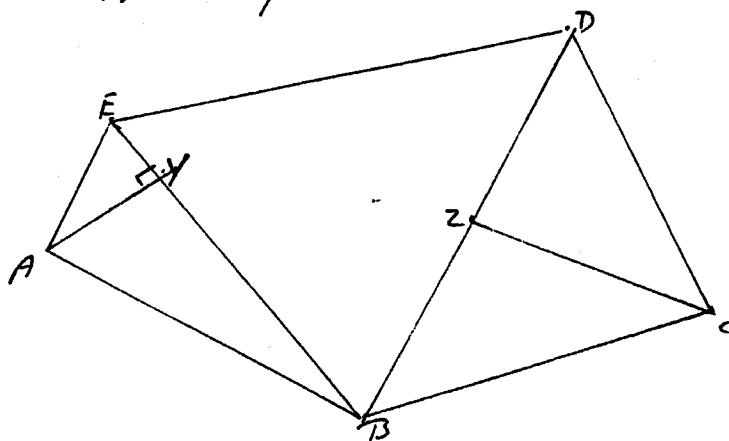
5b.

$$\triangle ABY \parallel \triangle AEB$$

(B is common)  
( $\hat{EAB} = \hat{EYA} = 90^\circ$ )

$$\frac{AB}{EB} = \frac{BY}{AB} \quad \left( \begin{array}{l} \text{Corresponding sides} \\ \text{in similar } \triangle's \end{array} \right)$$

$$\therefore AB^2 = BY \cdot BE$$



Q. In  $\triangle BEZ$ ,  $\triangle BDY$

$$\frac{BZ}{BY} = \frac{BE}{BD}$$

$$\left\{ \begin{array}{l} AB = BC \\ AB^2 = BY \cdot BE \\ BC^2 = BZ \cdot BD \end{array} \right.$$

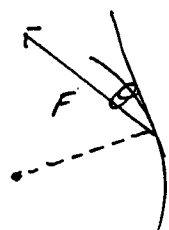
$\hat{B}$  is common.

$\therefore \triangle BEZ \parallel \triangle BDY$  -- (Two sides in the same ratio & included angle equal.)

(iii)  $\hat{BEY} = \hat{YDB}$  ... corresponding angles in similar triangles  $\triangle BEZ \parallel \triangle BDY$

$\therefore ZY$  subtends 2 equal angles at 2 points on the same side of it  $\therefore$  the end points ( $Z$  &  $Y$ ) and the 2 other points ( $E$  &  $D$ ) are concyclic.

Q6  
(a)



$$F \cos \theta = 600$$

$$F \sin \theta = \frac{mv}{r}$$

$$(i) \tan \theta = \frac{mv}{600r}$$

$$= \frac{300 \cdot 4\pi}{600 \cdot 60}$$

$$\theta \approx 18^\circ 13'$$

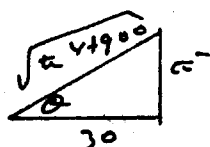
$$\therefore \theta \approx 18^\circ$$

$$(i) F \cos \theta = 600$$

$$F = \frac{600}{\cos \theta}$$

$$= 600 \cdot \sec \theta$$

$$= 631.6 \text{ N}$$



$$(ii) m\ddot{x} = p - mkv$$

$$(i) \ddot{x} = \frac{p}{m} - kv$$

$$\frac{dv}{dt} = \frac{p}{m} - kv$$

$$\int \frac{m dv}{p - mkv} = \int dt$$

$$\left[ -\frac{1}{k} \ln(p - mkv) \right]_2^4 = 5$$

$$\ln \frac{p - 4mk}{p - 2mk} = -5k$$

$$e^{5k} = \frac{p - 2mk}{p - 4mk}$$

$$p(1 - e^{5k}) = 2mk(1 - e^{5k})$$

2001 4U EXT 2 Trial

$$P = \frac{2km(2e^{5k} - 1)}{e^{5k} - 1}$$

$$(ii) v \frac{dv}{dx} = \frac{p}{m} - kv$$

$$\frac{mv}{p - mkv} dv = dx$$

$$\int_2^4 \left( -\frac{1}{k} + \frac{p}{k(p - mkv)} \right) dv = \int dx$$

$$\left[ -\frac{v}{k} + \frac{p}{mk} \ln(p - mkv) \right]_2^4 = x$$

$$-\frac{4}{k} - \frac{p}{mk} \ln(p - 4mk)$$

$$- \left( -\frac{2}{k} - \frac{p}{mk} \ln(p - 2mk) \right) = x$$

$$-\frac{2}{k} + \frac{p}{mk} \left( \ln \frac{p - 2mk}{p - 4mk} \right) = x$$

$$(iii) m\ddot{x} = p - mkv$$

for  $\ddot{x} > 0$  the train will continue to accelerate in the "positive" direction

$$i.e. p - mkv > 0$$

$$v < \frac{p}{mk}$$

If  $v = \frac{p}{mk}$   $\ddot{x} = 0$   $\therefore$  train is in equilibrium under the action of the 2 forces & thus will subsequently travel with uniform speed of  $\frac{p}{mk}$

$\therefore$  Upper bound is  $\frac{p}{mk}$ .

$$7/ (a) \quad (i) \quad m_p = \frac{ap - aq}{2ap - 2aq} = \frac{1}{2}$$

$$m_{oq} = \frac{q}{2}$$

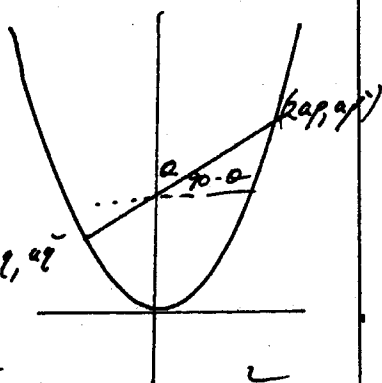
$$\therefore \frac{1}{2} \cdot \frac{q}{2} = -1 \quad \dots (\text{Since perp})$$

$$pq = -4$$

(iii)

$$\tan(90 - \theta) = \frac{ap - aq}{2ap - 2aq}$$

$$\cot \theta = \frac{p+q}{2} \quad 2\theta, q$$



$$d^2 = (ap - aq)^2 + (2ap - 2aq)^2$$

$$= a^2(p - q)^2 + 4a^2(p - q)^2$$

$$= a^2(p - q)^2(p + q)^2 + 4a^2(p - q)^2$$

$$= a^2(p - q)^2[(p + q)^2 + 4]$$

$$d = a(p - q)\sqrt{(p + q)^2 + 4}$$

$$= a(p - q)\sqrt{4\cot^2 \theta + 4}$$

$$= 2a(p - q)\csc \theta$$

$$\text{Now } (p + q)^2 = 4\cot^2 \theta$$

$$(p - q)^2 + 4pq = 4\cot^2 \theta$$

$$(p - q)^2 = 4\cot^2 \theta + 16 \quad pq = -4$$

$$= 2\sqrt{4\cot^2 \theta + 16}$$

$$\therefore d = 2a \csc \theta \cdot 2\sqrt{4\cot^2 \theta + 16}$$

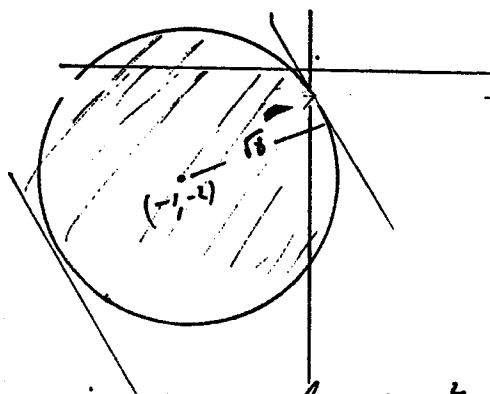
$$(b) \quad z\bar{z} + (1-2i)z + (1+2i)\bar{z} \leq 3$$

$$(x+iy)(x-iy) + (1-2i)(x+iy) + (1+2i)(x-iy) \leq 3$$

$$x^2 + y^2 + x + iy - 2ix + 2y + x - iy + 2ix - 2y \leq 3$$

$$x^2 + 2x + 1 + y^2 + 4y + 4 \leq 8$$

$$(x+1)^2 + (y+2)^2 \leq 8$$



Solving  $x+y=k$  &  $(x+1)^2 + (y+2)^2 = 8$

$$(x+1)^2 + (k-x)^2 + 4(k-x) + 4 = 8$$

$$x^2 + 2x + 1 + k^2 - 2kx + x^2 + 4k - 4x + 4 = 8$$

$$2x^2 + x(-2-2k) + k^2 + 4k - 3 = 0$$

$$\Delta = (-2-2k)^2 - 8(k^2 + 4k - 3) = 0$$

for Tangency

$$4 + 4k^2 + 8k - 8k^2 - 32k + 24 = 0$$

$$28 - 4k^2 - 24k = 0$$

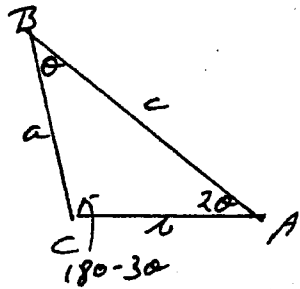
$$-4(k+7)(k-1) = 0$$

$$k = -7 \text{ or } 1$$

$$\therefore \text{Max} = 1$$

$$\text{Min} = -7$$

$$\begin{aligned}\sin 3\theta &= \sin(2\theta + \theta) \\ &= \sin 2\theta \cos \theta + \cos 2\theta \sin \theta \\ &= 2\sin \theta \cos^2 \theta + (\cos^2 \theta - \sin^2 \theta) \sin \theta \\ &= 3\cos^2 \theta \sin \theta - \sin^3 \theta \\ &= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta \\ &= 3\sin \theta - 4\sin^3 \theta.\end{aligned}$$



Using sine rule

$$\frac{a}{\sin \theta \cos \theta} = \frac{c}{3\sin \theta - 4\sin^3 \theta} = \frac{b}{\sin \theta}$$

$$a = \frac{a(3\sin \theta - 4\sin^3 \theta)}{2\sin \theta \cos \theta}; b = \frac{a \sin \theta}{2\sin \theta \cos \theta} = \frac{a}{2\cos \theta}$$

$$\begin{aligned}b(b+c) &= \frac{a}{2\cos \theta} \left( \frac{a}{2\cos \theta} + \frac{a(3\sin \theta - 4\sin^3 \theta)}{2\sin \theta \cos \theta} \right) \\ &= \frac{a^2}{2\cos \theta} \left( \frac{4(1 - \sin^2 \theta)}{2\cos \theta} \right) \\ &= \frac{a^2}{2\cos \theta} (2 \cdot \cos \theta) \\ &= a^2\end{aligned}$$

ii) Using cos rule

$$\cos \beta = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\begin{aligned}2\cos \beta &= \frac{b+c}{a} \dots a^2 = b^2 + bc \\ &= \frac{b^2 + bc}{ab} \\ &= \frac{a}{1} \dots a^2 = b^2 + bc\end{aligned}$$

(iii)  $\frac{n}{m} = \frac{a}{b}$

Let  $a = ln$ ,  $b = lm$

Now  $a^2 = b(b+c)$

$$l^2 n^2 = lm(lm+c)$$

$$\therefore c = \frac{ln^2}{m} - lm$$

but  $c$  is integral &  $n \times m$  have no common factor

$$\therefore \text{let } \frac{ln}{m} = k$$

$$\begin{aligned}\text{Then } b &= lm & b+c &= lm + \frac{ln^2}{m} - lm \\ &= m^2 k & &= m^2 k + \frac{mkn \cdot mk}{m} \\ & & &= kn^2\end{aligned}$$

(iv) From diagram  $0 < \beta < 30^\circ$

$$\therefore \frac{\sqrt{3}}{2} < \cos \beta < 1$$

$$\therefore \sqrt{3} < \frac{n}{m} < 2 \dots 2\cos \beta = \frac{n}{m}$$

$$\sqrt{3}m < n < 2m$$

inspection smallest integral values of  $n \times m$  are 7 & 4 respectively.

(v)  $b = km^2$ ,  $b+c = kn^2$   
 $\times a = b(b+c)$

$$\begin{aligned}b &= 16k, c = 32k, a = 28k \\ \text{Hence smallest } a \text{ with integral sides is } 16, 28, 32\end{aligned}$$

$$\therefore \text{This Problem} = 77$$